

B.C.A. Ist Semester Examination,

February-2022

MATHEMATICS

(Paper : BCA-103)

Time allowed : 3 hours]

[Maximum marks : 80

Note: Attempt five questions in all. Question No. 1 is compulsory and attempt four more questions by selecting one question from each unit. All questions carry equal marks.

1. (i) Write down all the subsets of - $8 \times 2 = 16$
 $\{3, 0, \pi\}$

- (ii) If $\begin{bmatrix} x+3 & x+2y \\ z-1 & 4u-6 \end{bmatrix} = \begin{bmatrix} 0 & -7 \\ 3 & 2u \end{bmatrix}$, find the values
 x, y, z and u . <https://www.pupilstutor.com/>

- (iii) Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

- (iv) Find the domain of $y = \sqrt{x+5}$

- (v) Find dy/dx , when $y = \frac{x^2 + x + 1}{x + 5}$

- (vi) Find dy/dx , when $y = a^{3x-10}$

- (vii) Evaluate $\int \left(x + \frac{1}{x}\right)^2 dx$

- (viii) Evaluate $\int \cos^2 x dx$

Unit-I

2. (i) If $A = \{x : x = 2^n, n \leq 5, n \in \mathbb{N}\}$ and 8

$$B = \{x : x = 4^n, n \leq 4, n \in \mathbb{N}\}; \text{ find}$$

- (a) $A \cup B$ (b) $A \cap B$ (c) $A - B$ (d) $B - A$

- (ii) If $A = \begin{bmatrix} 0 & 3 \\ -2 & 5 \end{bmatrix}$, then find 'k' so that $kA^2 = 5A - 6I$,

Where I is a unit matrix. 8

3. (i) Prove that $\begin{bmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{bmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$

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- (ii) Solve the following system of equations using matrices : 8

$$x - y - z = 1; 2x + y + z = 2; x - 2y + z = 4$$

Unit-II

4. (i) Show that the relation $R = \{(a, b) \in \mathbb{R} : a + b \text{ is even}\}$ on the set of positive integers is not a partial order relation but is an equivalence relation. 8

- (ii) If $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = \frac{5x+3}{7}$, $x \in \mathbb{R}$;
 prove that f is bijective function and hence find f^{-1} .

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5. (i) Evaluate the limit: $\lim_{x \rightarrow \pi/4} \left(\frac{\sin x - \cos x}{x - \pi/4} \right)$ 8

- (ii) (a) If $f(x) = \begin{cases} \frac{\sin 2x}{\sin 3x}, & \text{when } x \neq 0 \\ 2, & \text{when } x = 0 \end{cases}$; find whether

$f(x)$ is continuous at $x = 0$.

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- (b) Determine the value of 'k' for which the following function (x) is continuous at $x = 3$:

$$f(x) = \begin{cases} \frac{(x+3)^2 - 36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

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Unit-III

6. (i) If $y = x + \sqrt{x^2 - 1}$, Prove that $(y-x) \frac{dy}{dx} - y = 0$

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- (ii) Find $\frac{dy}{dx}$, if $y = \frac{\sqrt{x-1}}{\sqrt{x+1}}$

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(iii) Find $\frac{dy}{dx}$, if $y = \cot^{-1} (\operatorname{cosec} x + \cot x)$

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(iv) Find $\frac{dy}{dx}$, if $y = \tan^{-1} \left(\frac{\sqrt{1+x^2} + 1}{x} \right)$

4

7. (i) Find $\frac{dy}{dx}$, if $y = x^x + (\log x)^x$

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(ii) Differentiate $\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$ w.r.t. $\tan^{-1} x$.

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Unit-IV

8. Evaluate the following integral

(i) $\int \frac{\tan x}{\sec x + \tan x} dx$

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(ii) $\int \tan^{-1} x dx$

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9. Evaluate the following integral

(i) $\int \frac{3x+2}{(x+2)(x+3)} dx$

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(ii) $\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x} dx$

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